

Birthday attack

Module IV



Introduction

- Imagine a group of people, how big do you think the group would have to be before there is more than a 50% chance that two people in the group have the same birthday?
- (Assume, there are no twins, every birthday is equally likely, and ignore leap years.)
- The answer seems to be very low.



In a group of 23 people, there is a 50.73% chance that two people will share the same birthday

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- But with 365 days in a year, How's this possible that you need such a small group to get even odds of a shared birthday?
- Why is our Intuition so wrong?
- To understand this, use combinatorics, which deals with the likelihoods of different combinations.

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- Trying to calculate the odds of a match directly is challenging, since there are many ways to find a birthday match in a group.
- Instead, it's easier to calculate the odds that everyone's birthday is different.
- Even there is a birthday match in a group, so the odds of a match and the odds of a no match must add up to 100%. It means, we can find the probability of a match by subtracting a no match from 100.

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- Calculate the odds that just one pair of people have different birthdays.
- One day of the year will be person A's birthday, which leaves only 364 possible birthdays for person B.
- The probability of different birthdays for A & B or any pair of people, is $364/365 = 0.997$ or 99.7 % very high.
- Accommodate person C in the small group, the probability of she is having different birthday is $363/365$.
- And so on.

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- $$\frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdot \frac{362}{365} \cdot \frac{361}{365} \cdot \frac{360}{365} \cdot \frac{359}{365} \cdot \frac{358}{365} \cdot \frac{357}{365} \cdot \frac{356}{365} \cdot \frac{355}{365} \cdot \frac{354}{365} \cdot \frac{353}{365} \cdot \frac{352}{365} \cdot \frac{351}{365} \cdot \frac{350}{365} \cdot \frac{349}{365} \cdot \frac{348}{365} \cdot \frac{347}{365} \cdot \frac{346}{365} \cdot \frac{345}{365} \cdot \frac{344}{365} \cdot \frac{343}{365}$$

- The probability that no one shares a birthday = 0.4927.
- So, there is a 49.27% chance that no one in the group of 23 people shares a birthday.
- Then, if the value is subtracted from 100 then, we get 50.73% chance of at-least one birthday match, better than even odds.
- The key to such a high probability of a match in a relatively small group is the surprisingly large number of possible pairs.
- As a group grows, the number of possible combinations gets bigger much faster.

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- A group of five people ten possible pairs. ($5*4 / 2 = 10$).
- A group of 10 people have $10*9/2=45$ pairs.
- A group of 23 people has $23*22/2=253$ pairs. Everyone has a chance of getting same birthday.
- The number of pairs grows quadratically, meaning that it is proportional to the square of the number of people in the group.